

Elastic bending stress verification for a simply supported rectangular timber beam under uniform load.

1. Input Parameters

Symbol	Value	Unit	Description
b	200	mm	Section width
h	400	mm	Section depth
L	5000	mm	Span
q	4.5	kN/m	Distributed load (design)
f_m	24	MPa	Design bending strength

2. Calculation

Maximum bending moment

$$\begin{aligned}
 M_{max} &= \frac{q \cdot \left(\frac{L}{1000}\right)^2}{8} \\
 &= \frac{4.5 \cdot \left(\frac{5000}{1000}\right)^2}{8} \\
 &= \mathbf{14.06 \text{ kNm}}
 \end{aligned}$$

Elastic section modulus

$$\begin{aligned}
 W &= \frac{b \cdot h^2}{6} \\
 &= \frac{200 \cdot 400^2}{6} \\
 &= \mathbf{5.333e + 06 \text{ mm}^3}
 \end{aligned}$$

Maximum bending stress

$$\begin{aligned}
 \sigma_m &= M_{max} \cdot 1000000 \cdot \frac{1}{W} \\
 &= 14.06 \cdot 1000000 \cdot \frac{1}{5.333e + 06} \\
 &= \mathbf{2.637 \text{ MPa}}
 \end{aligned}$$

3. Verification

OK $\sigma_m = 2.637 \text{ MPa} \leq f_m = 24 \text{ MPa}$